# KEYS IN DBMS

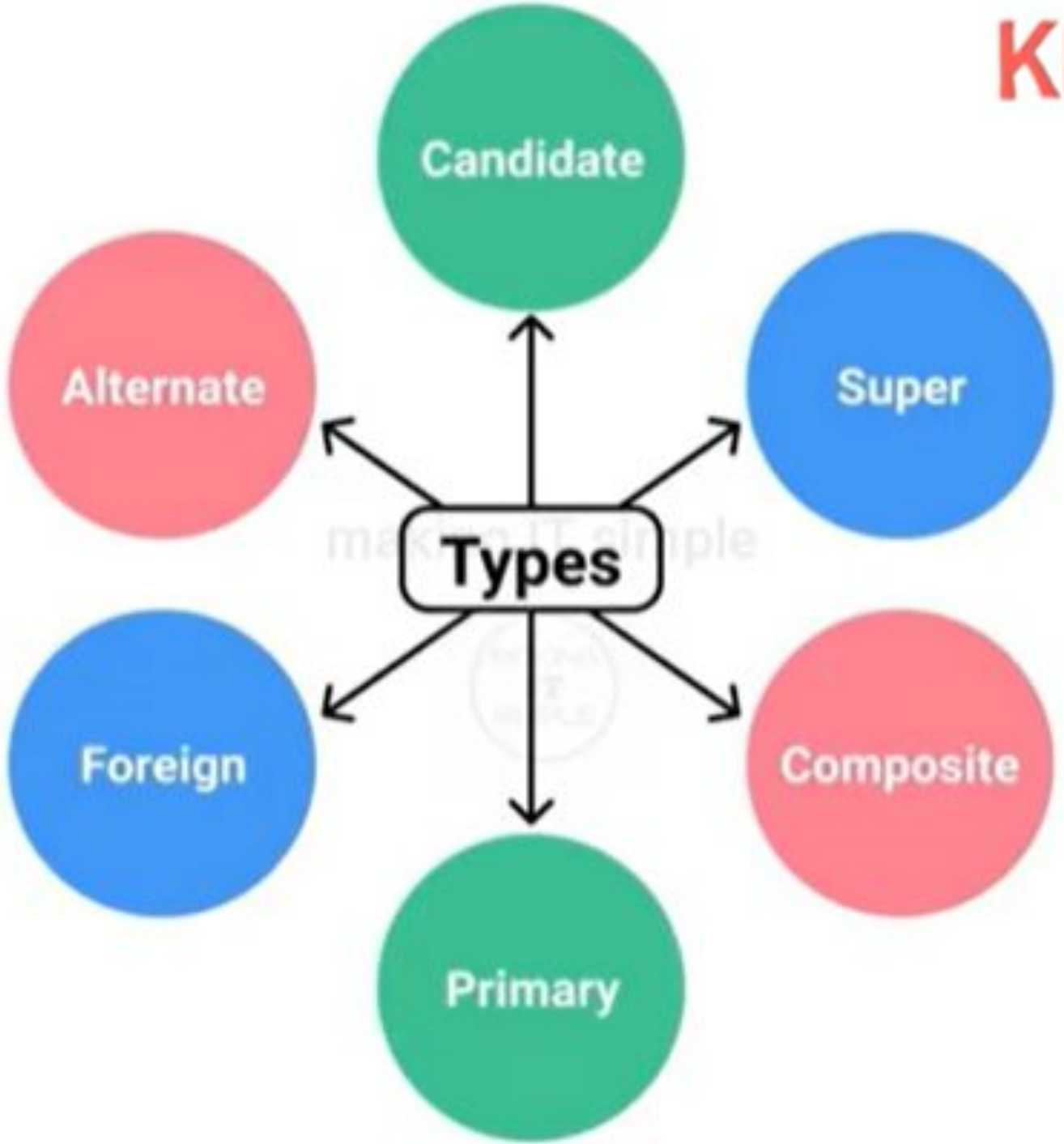
# KEYS IN DBMS

| ID | Vendor | Name | Number | Date | Travel Da | Trip Total |
|---|---|---|---|---|---|---|
| 1 | Onxy | Connolly | 998009af | 01/05/16 | 11/25/16 | $ 15,000.00 |
| 2 | Regent | Cummings | asdfbasdfb | 04/15/16 | 11/25/16 | $ 25,000.00 |
| 3 | Caravan Tours | Daly | 555999a | 05/05/16 | 10/01/16 | $ 20,000.00 |
| 4 | Alexander & Roberts | Smitha | 12ab | 05/15/16 | 11/25/16 | $ 32,000.00 |
| 5 | Rocky Mountaineer | Washington | c99df999 | 05/15/16 | 11/25/16 | $ 25,000.00 |
| 6 | Celestyal Cruises | Jones | 3fag | 06/01/16 | 12/25/16 | $ 32,000.00 |
| 7 | Crystal River Cruises | Hamilton | a0a0a0 | 06/06/16 | 01/01/17 | $ 25,000.00 |
| 8 | Rocky Mountaineer | Kelly | bcde490 | 07/07/16 | 01/05/17 | $ 40,000.00 |
| 9 | Uniworld | Jones | 88888855 | 07/07/16 | 12/25/16 | $ 15,000.00 |
| 10 | Cunard | Browning | 56575859 | 08/08/16 | 12/01/16 | $ 17,999.00 |
| 11 | Rail Europe | Quinn | 4jkjhaa | 08/08/16 | 12/25/16 | $ 20,000.00 |
| 12 | Hotel Rewards | Taylor | 1111100000 | 09/09/16 | 12/03/16 | $ 8,000.00 |
| 13 | Carnival | Martin | b123333 | 10/10/16 | 12/03/16 | $ 8,000.00 |
| 14 | Celestyal Cruises | Clown | def | 11/11/16 | 02/14/17 | $ 15,000.00 |





# KEYS IN DBMS


Candidate
Alternate
Super
Types
Foreign
Composite
Primary

# KEYS IN DBMS

**DEFINITION:-** It is an attribute or set of attributes that help to uniquely identify a tuple or row in a table.




| Roll No | Name | Age |
|---|---|---|
| 1 | ABC | 18 |
| 2 | PQR | 17 |
| 3 | MNO | 19 |
| 4 | PQR | 18 |

**Table**: student

# KEYS IN DBMS

AGE OF ALL STUDENTS?

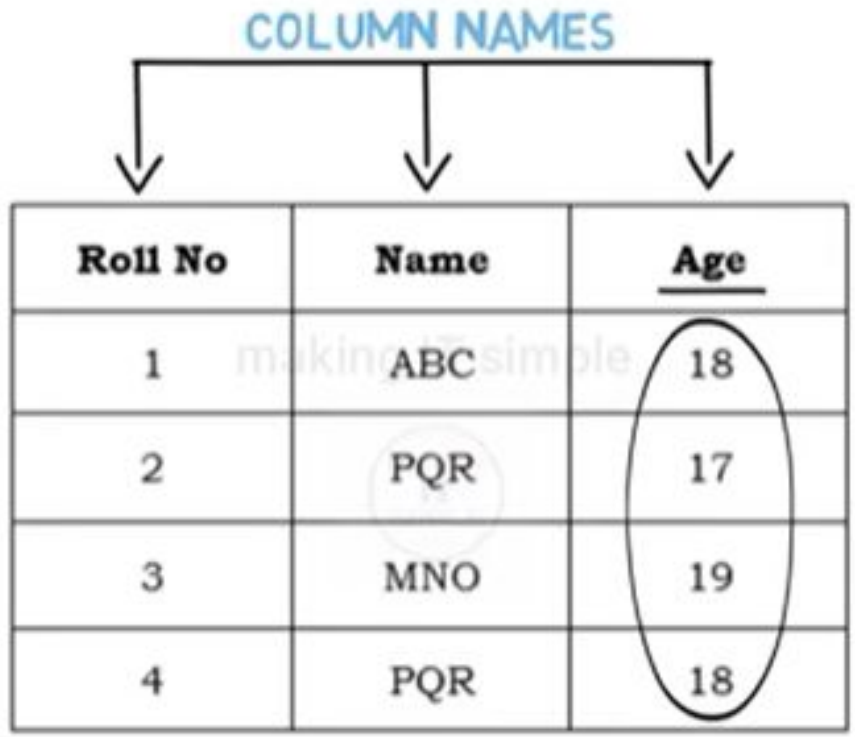
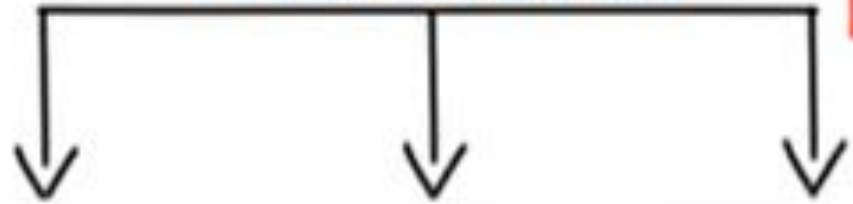



**Table**: student

# KEYS IN DBMS

COLUMN NAMES



| Roll No | Name | Age |
|---|---|---|
| 1 | ABC | 18 |
| 2 | PQR | 17 |
| 3 | MNO | 19 |
| 4 | PQR | 18 |

**Table**: student

NO NAMES TO UNIQUELY IDENTIFY ROWS

UNIQUE ROLL NO

CAN BE USED TO UNIQUELY IDENTIFY UNIQUE ROWS

# KEYS IN DBMS

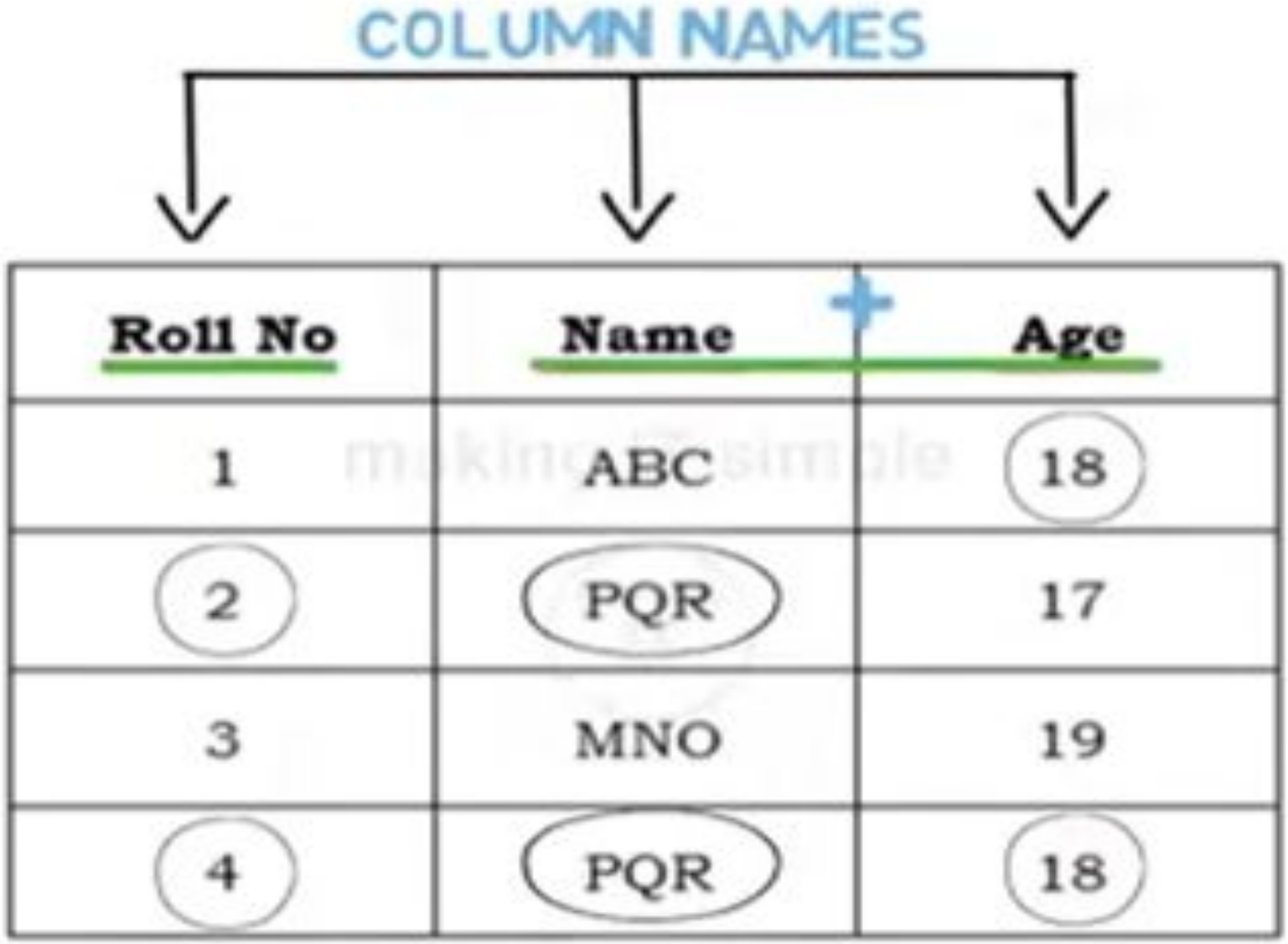



**Table**: student

ABC · 18

PQR · 17

MNO · 19

PQR · 18

# KEYS IN DBMS

(ROLL NO) : KEY
SINGLE ATTRIBUTE

(NAME , AGE) : KEY
SET OF ATTRIBUTES

DEFINITION:- It is an attribute or set of attributes that help to uniquely identify a tuple or row in a table.

# PRIMARY KEY

# PRIMARY KEY

## DEFINITION:-

Attribute or group of attributes of a table must

a) contain unique value for each row

b) cannot contain null values

then only it can be considered as a primary key.

| Registration Number | Roll No | Name | Age | Mobile Number | Bank Acc. Number |
|---|---|---|---|---|---|
| REG001 | 1 | ABC | 18 | 1005487 | BAM0256789 |
| REG002 | 2 | PQR | 17 | 4536787 | BAM2355765 |
| REG003 | 3 | MNO | 19 | 6895584 | BAM2276854 |
| REG004 | 4 | PQR | 18 | 5487597 | BAM9876332 |

**Table**: student

# PRIMARY KEY

| Registration Number | Roll No | Name | Age | Mobile Number | Bank Acc. Number |
|---|---|---|---|---|---|
| REG001 | 1 | ABC | 18 | 1005487 | BAM0256789 |
| REG002 | 2 | PQR | 17 | 4536787 | BAM2355765 |
| REG003 | 3 | MNO | 19 | 6895584 | BAM2276854 |
| REG004 | 4 | PQR | 18 | 5487597 | BAM9876332 |

| COLUMN NAME | UNIQUE VALUES | NO NULL VALUES | PRIMARY KEY |
|---|---|---|---|

# PRIMARY KEY

| Registration Number | Roll No | Name | Age | Mobile Number | Bank Acc. Numbe |
|---|---|---|---|---|---|
| REG001 | 1 | ABC | 18 | 1005487 | BAM0256789 |
| REG002 | 2 | PQR | 17 | 4536787 | BAM2355765 |
| REG003 | 3 | MNO | 19 | 6895584 | BAM2276854 |
| REG004 | 4 | PQR | 18 | 5487597 | BAM9876332 |

**Table**: student



# SUPER KEY

# SUPER KEY

**SUPER KEY** - It is an attribute or set of attributes that is used to uniquely identify all attributes in a relation or a table.

Primary Key

| Roll No. | Name | Age |
|---|---|---|
| 1 | ABC | 18 |
| 2 | PQR | 21 |
| 3 | MNO | 22 |
| 4 | SDK | 20 |
| 5 | PQR | 19 |
| 6 | URL | 20 |
| 7 | ABC | 18 |

Table: Student

Primary Key - **UNIQUE AND NOT NULL**

making IT simple

# SUPER KEY

**SUPER KEY** - It is an attribute or set of attributes that is used to uniquely identify all attributes in a relation or a table.

Primary Key - UNIQUE AND NOT NULL

| Roll No. | Name | Age |
|---|---|---|
| 1 | ABC | 18 |
| 2 | PQR | 21 |
| 3 | MNO | 22 |
| 4 | SDK | 20 |
| 5 | PQR | 19 |
| 6 | URL | 20 |
| 7 | ABC | 18 |

Table: Student

ROLL NO ⟶ NAME, AGE ✓

NAME ⟶ ✗ ROLL NO., AGE

AGE ⟶ ✗

ROLL NO., NAME ⟶ AGE

ROLL NO., AGE ⟶ NAME

NAME, AGE ✗

ROLL NO., NAME, AGE

| Roll No. | Name | Age |
|---|---|---|
| 1 | ABC | 18 |
| 1 | ABC | 18 |

Super Key = {Roll No., Roll No. Name , Roll No. Age , Roll No. Name Age }

# SUPER KEY

DIFFERENCE Super Keys = { Roll No. , Roll No. Name , Roll No. Age , Roll No. Name Age }

PRIMARY KEY (ROLL NO.) ' SUPER KEY

ALL PRIMARY KEYS ARE SUPER KEYS BUT NOT ALL SUPER KEYS ARE PRIMARY KEYS

SUPER KEYS ' MULTIPLE | PRIMARY KEY ' SINGLE

# CANDIDATE KEY

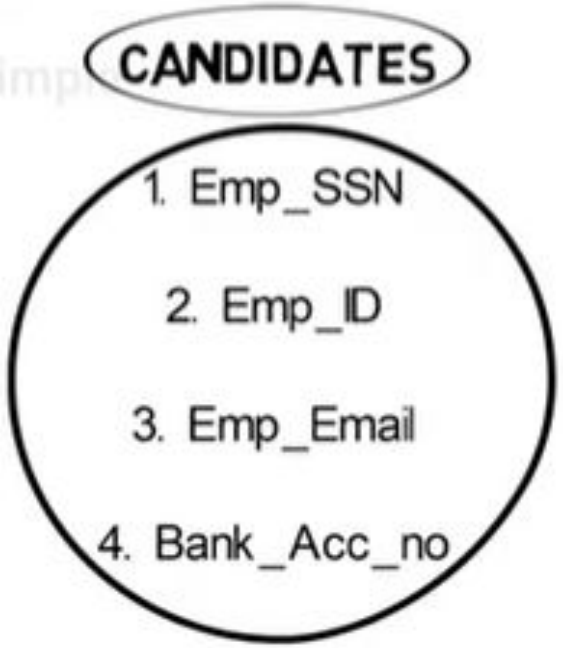
# CANDIDATE KEY

## Primary Key

| Emp_SSN | Emp_ID | Emp_Name | Emp_Email | Bank_Acc_No |
|---|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com | 374673563978 |
| 124 | 6 | DEF | def@yt.com | 875642565348 |
| 125 | 8 | GHI | ghi@yt.com | 435763462879 |
| 126 | 10 | JKL | jkl@yt.com | 278554897345 |

1. Must contain unique values

2. Null values not allowed




EMP_SSN

PRIMARY KEY

# CANDIDATE KEY

## DEFINITION

Candidate key is a super key whose no proper subset is a super key.

The minimal set of attribute that can uniquely identify a tuple is known as candidate key.

A minimal super key is called candidate key.

# CANDIDATE KEY

| Reg_No | Roll_No | Student_Name | Student_Email |
|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com |
| 124 | 6 | DEF | def@yt.com |
| 125 | 8 | GHI | ghi@yt.com |
| 126 | 10 | JKL | jkl@yt.com |

## DEFINITION

Candidate key is a super key whose no proper subset is a super key.

The minimal set of attribute that can uniquely identify a tuple is known as candidate key.

A minimal super key is called candidate key.

Super Keys = { Reg_No },
{ Roll_No },
{ Student_Email },
{ Reg_No, Roll_No },
{ Roll_No, Student_Name },
{ Reg_No, Roll_No, Student_Email },
{ Reg_No, Roll_No, Student_Name }

# CANDIDATE KEY

## PROPER SUBSET ???

A subset that excludes at least one member of the containing set.

$\{2, 4, 6\}$
Super Key

$\{2\}, \{4\}, \{6\}, \{2, 4\}, \{2, 6\}, \{4, 6\}$
Proper Subsets

# CANDIDATE KEY

| Reg_No | Roll_No | Student_Name | Student_Email |
|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com |
| 124 | 6 | DEF | def@yt.com |
| 125 | 8 | GHI | ghi@yt.com |
| 126 | 10 | JKL | jkl@yt.com |

Super Keys = { Reg_No },
{ Roll_No },
{ Student_Email },
{ Reg_No, Roll_No },
{ Roll_No, Student_Name },
{ Reg_No, Roll_No, Student_Email },
{ Reg_No, Roll_No, Student_Name }

## DEFINITION

CANDIDATE SUPER KEY PROPER SUBSETS

Candidate key is a super key whose no proper subset is a super key.

The minimal set of attribute that can uniquely identify a tuple is known as candidate key.

A minimal super key is called candidate key.

making IT simple

# CANDIDATE KEY

{ Reg_No }

PROPER SUBSETS

{ }

{ Roll_No },

PROPER SUBSETS

{ }

{ Student_Email },

PROPER SUBSETS

{ }

{ Reg_No, Roll_No },

PROPER SUBSETS

{ Reg_No }

{ Roll_No }

{ Roll_No, Student_Name },

PROPER SUBSETS

{ Roll_No }

{ Student_Name }

{ Reg_No, Roll_No, Student_Email },

PROPER SUBSETS

{ Reg_No } S.K.
{ Roll_No } S.K.
{ Student_Email } S.K.
{ Reg_No, Roll_No }
{ Reg_No, Student_Email }
{ Roll_No, Student_Email }

{ Reg_No, Student_Name, Roll_No }

PROPER SUBSETS

{ Reg_No } S.K.
{ Roll_No } S.K.
{ Student_Name }
{ Reg_No, Roll_No }
{ Reg_No, Student_Name }
{ Roll_No, Student_Name }

# CANDIDATE KEY

| Reg_No | Roll_No | Student_Name | Student_Email |
|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com |
| 124 | 6 | DEF | def@yt.com |
| 125 | 8 | GHI | ghi@yt.com |
| 126 | 10 | JKL | jkl@yt.com |

Super Keys = { Reg_No },
{ Roll_No },
{ Student_Email },
{ Reg_No, Roll_No },
{ Roll_No, Student_Name },
{ Reg_No, Roll_No, Student_Email },
{ Reg_No, Roll_No, Student_Name }

## DEFINITION

Candidate key is a super key whose no proper subset is a super key.

The minimal set of attribute that can uniquely identify a tuple is known as candidate key.

A minimal super key is called candidate key.

| CANDIDATE | SUPER KEY | PROPER SUBSETS |
|---|---|---|
| 👍 | { Reg_No } | { Reg_No } S.K. |
| 👍 | { Roll_No } | { Roll_No } S.K. |
| 👍 | { Student_Email } | { Student_Name } |
| | { Reg_No, Roll_No } | { Reg_No, Roll_No } |
| | { Roll_No, Student_Name } | { Reg_No, Student_Name } |
| | { Reg_No, Roll_No, Student_Email } | { Roll_No, Student_Name } |
| | { Reg_No, Student_Name, Roll_No } | |

# CANDIDATE KEY

| Reg_No | Roll_No | Student_Name | Student_Email |
|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com |
| 124 | 6 | DEF | def@yt.com |
| 125 | 8 | GHI | ghi@yt.com |
| 126 | 10 | JKL | jkl@yt.com |

Candidate Keys = { Reg_No },
{ Roll_No },
{ Student_Email }

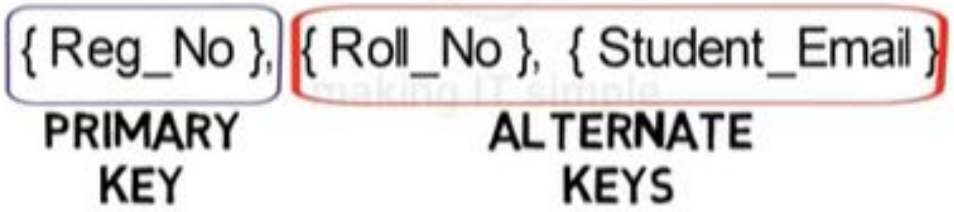
**In Candidate Keys, Super keys with unnecessary attributes are ignored**

# ALTERNATE KEYS

# ALTERNATE KEYS

| Reg_No | Roll_No | Student_Name | Student_Email |
|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com |
| 124 | 6 | DEF | def@yt.com |
| 125 | 8 | GHI | ghi@yt.com |
| 126 | 10 | JKL | jkl@yt.com |

Candidate Keys = { Reg_No },
{ Roll_No },
{ Student_Email }




**Among the Candidate Keys, one is selected as the Primary key and all others are called as Alternate keys**

# ALTERNATE KEYS

## ALTERNATE KEY

- RELATION WITH PRIMARY & CANDIDATE KEY

P.K.

| Emp_SSN | Emp_ID | Emp_Name | Emp_Email | Bank_Acc_No |
|---|---|---|---|---|
| 123 | 3 | ABC | abc@yt.com | 374673563978 |
| 124 | 6 | DEF | def@yt.com | 875642565348 |
| 125 | 8 | GHI | ghi@yt.com | 435763462879 |
| 126 | 10 | JKL | jkl@yt.com | 278554897345 |

CANDIDATE KEYS IN FOLLOWING TABLE:

EMP_SSN | EMP_ID , EMP_EMAIL , BANK_ACC_NO

P.K. ALTERNATE KEYS

1) UNIQUE VALUES 2) NOT NULL VALUES

DEFINITION:- ALTERNATE KEY IS A KEY THAT HAS NOT BEEN SELECTED TO BE THE PRIMARY KEY, BUT IS A CANDIDATE KEY.

# FOREIGN KEY

FOREIGN KEY

EMPLOYEES

< RELATION >

DEPARTMENTS

IS WORKING

P.K.

| Emp_Id | Emp_Name | Emp_Email |
|---|---|---|
| 12 | H | h@mail.com |
| 23 | I | i@mail.com |
| 14 | J | j@mail.com |
| 33 | K | k@mail.com |
| 37 | L | l@mail.com |
| 27 | M | m@mail.com |

?

P.K.

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

**The details of one table should be included in another table**

EMPLOYEES

< RELATION >

IS WORKING

DEPARTMENTS

P.K.

| Emp_Id | Emp_Name | Emp_Email |
|---|---|---|
| 12 | H | h@mail.com |
| 23 | I | i@mail.com |
| 14 | J | j@mail.com |
| 33 | K | k@mail.com |
| 37 | L | l@mail.com |
| 27 | M | m@mail.com |




P.K.

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

**Adding Employee details in the Department table.**

**One Department may have so many employees. So it is not possible.**

| Dept_Id | Dept_Name | Dept_Head | Emp |
|---|---|---|---|
| 1 | Research | R | 12, 14 |
| 2 | Manufacturing | M | 23, 27 |
| 3 | Production | P | 33 |
| 4 | Distribution | D | 37 |

# FOREIGN KEY

EMPLOYEES

< RELATION >

IS WORKING

DEPARTMENTS

P.K.

| Emp_Id | Emp_Name | Emp_Email |
|---|---|---|
| 12 | H | h@mail.com |
| 23 | I | i@mail.com |
| 14 | J | j@mail.com |
| 33 | K | k@mail.com |
| 37 | L | l@mail.com |
| 27 | M | m@mail.com |

?

P.K.

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

**Department details are added in the Employee table. This is the optimal solution.**

**But is it necessary to add all the department details in the Employees table? No**

| Emp_Id | Emp_Name | Emp_Email | Dept_Id | Dept_Name | Dept_Head |
|---|---|---|---|---|---|
| 12 | H | h@mail.com | 1 | Research | R |
| 23 | I | i@mail.com | 2 | Manufacturing | M |
| 14 | J | j@mail.com | 1 | Research | R |
| 33 | K | k@mail.com | 3 | Production | P |
| 37 | L | l@mail.com | 4 | Distribution | D |
| 27 | M | m@mail.com | 2 | Manufacturing | M |

EMPLOYEES

< RELATION >

IS WORKING

DEPARTMENTS

P.K.

| Emp_Id | Emp_Name | Emp_Email |
|---|---|---|
| 12 | H | h@mail.com |
| 23 | I | i@mail.com |
| 14 | J | j@mail.com |
| 33 | K | k@mail.com |
| 37 | L | l@mail.com |
| 27 | M | m@mail.com |




P.K.

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

**It is enough only to add the Primary Key of the Department table into the Employee table.**

EMPLOYEE K

IS WORKING IN

PRODUCTION DEPARTMENT

DEPT_ID = 4

| Emp_Id | Emp_Name | Emp_Email | Dept_Id |
|---|---|---|---|
| 12 | H | h@mail.com | 1 |
| 23 | I | i@mail.com | 2 |
| 14 | J | j@mail.com | 1 |
| 33 | K | k@mail.com | 3 |
| 37 | L | l@mail.com | 4 |
| 27 | M | m@mail.com | 2 |

# FOREIGN KEY

EMPLOYEES

< RELATION >

IS WORKING

DEPARTMENTS

P.K.

| Emp_Id | Emp_Name | Emp_Email |
|---|---|---|
| 12 | H | h@mail.com |
| 23 | I | i@mail.com |
| 14 | J | j@mail.com |
| 33 | K | k@mail.com |
| 37 | L | l@mail.com |
| 27 | M | m@mail.com |

?

P.K.

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

DEPT_ID = 4

FOREIGN KEY

| Emp_Id | Emp_Name | Emp_Email | Dept_Id |
|---|---|---|---|
| 12 | H | h@mail.com | 1 |
| 23 | I | i@mail.com | 2 |
| 14 | J | j@mail.com | 1 |
| 33 | K | k@mail.com | 3 |
| 37 | L | l@mail.com | 4 |
| 27 | M | m@mail.com | 2 |

# FOREIGN KEY

An attribute or set of attributes in a relational database table, that provides a link between two tables is called a foreign key.

It refers to the field in a table which is the primary key of another table.

More than one foreign keys are allowed in a table.

Foreign key attribute can contain duplicate values.

Foreign key maintains referential integrity in database.

P.K DEPARTMENT

| Dept_Id | Dept_Name | Dept_Head |
|---|---|---|
| 1 | Research | R |
| 2 | Manufacturing | M |
| 3 | Production | P |
| 4 | Distribution | D |

EMPLOYEE F.K F.K

| Emp_Id | Emp_Name | Emp_Email | Dept_Id | Proj_Id |
|---|---|---|---|---|
| 12 | H | h@mail.com | 1 | 151 |
| 23 | I | i@mail.com | 2 | 101 |
| 14 | J | j@mail.com | 1 | 101 |
| 33 | K | k@mail.com | 3 | 151 |
| 37 | L | l@mail.com | 4 | 101 |
| 27 | M | m@mail.com | 2 | 151 |

P.K

| Proj_Id | Proj_Name |
|---|---|
| 101 | ONM |
| 151 | RQP |

PROJECT

# COMPOSITE KEY

## COMPOSITE KEY

| Roll_No | Student_Name | Student_Age |
|---|---|---|
| 1 | A | 20 |
| 2 | B | 20 |
| 3 | C | 19 |
| 4 | D | 20 |
| 5 | E | 19 |

making IT simple

CAN YOU IDENTIFY UNIQUE RECORDS FROM THIS TABLE JUST BY USING A SINGLE ATTRIBUTE ?

YES , USING ROLL NUMBER ATTRIBUTE

ROLL NUMBER > PRIMARY KEY

## COMPOSITE KEY

| Roll_No | Student_Name | Subject_Name | Marks |
|---|---|---|---|
| 1 | A | Operating System | 80 |
| 1 | A | Database | 74 |
| 2 | B | Cloud Computing | 55 |
| 2 | B | Operating System | 60 |
| 3 | C | Database | 90 |
| 3 | C | Cloud Computing | 86 |
| 4 | D | Operating System | 79 |
| 4 | D | Database | 83 |

CAN YOU IDENTIFY UNIQUE RECORDS FROM THIS TABLE JUST BY USING A SINGLE ATTRIBUTE ?

IN THIS CASE ANSWER IS NO

# COMPOSITE KEY

| Roll_No | Student_Name | Subject_Name | Marks |
|---|---|---|---|
| 1 | A | Operating System | 80 |
| 1 | A | Database | 74 |
| 2 | B | Cloud Computing | 55 |
| 2 | B | Operating System | 60 |
| 3 | C | Database | 90 |
| 3 | C | Cloud Computing | 86 |
| 4 | D | Operating System | 79 |
| 4 | D | Database | 83 |

COMBINATION OF MORE THAN ONE ATTRIBUTE

( ROLL_NO , SUBJECT_NAME ) >>> ( 4 , OPERATING SYSTEM )
( 2 , CLOUD COMPUTING )

COMPOSITE KEY

**DEFINITION:** COMBINATION OF TWO OR MORE ATTRIBUTES USED TO UNIQUELY IDENTIFY RECORDS N A RELATION OR TABLE CALLED AS COMPOSITE KEY.